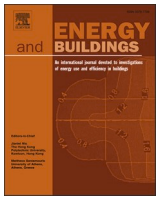
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Who pays more? An assessment of housing and population characteristics associated with energy costs

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ABSTRACT

The introduction of the EU ETS system in the residential building sector in times of energy crisis posits a risk of increased energy poverty. Monitoring energy poverty and the distributional impacts of energy consumption by the governing bodies is important for facilitating just energy policies and reducing inequalities. To evaluate the patterns of varied energy costs, the study compares different classification and regression algorithms by applying publicly available data to significantly increase the total number of buildings assessed in the study. The outputs of the machine learning algorithm are further used in association with the data on housing and spatial characteristics, as well as neighbourhood-level data on the occupational, ethnic, and demographic characteristics, to evaluate the socio-spatial variability of energy costs. The results reveal lower dwelling energy costs in housing estates erected during the '70 s to the '90 s. Furthermore, lower energy consumption was associated with higher shares of middle occupational groups, individuals whose mother tongue was not Estonian, and the elderly population. On the contrary, the inner city of Tallinn and outer city detached housing neighbourhoods with a higher share of the pre-WWII housing stock were associated with the city's highest energy costs–those areas dominated by the Estonian-speaking population and overall higher socio-economic status. This patterning illuminates how the differentiated exposure to energy costs leads to the new socio-spatial inequality that could be considered when planning financial incentives to advance energy retrofitting as well as alleviating the risks of increased energy poverty.

1. Introduction

The building sector accounts for 40 % of the final energy consumption and 36 % of the energy-related greenhouse gas emissions [1]. The Renovation Wave envisions a framework to decarbonize building stock with a strong focus on the accessibility of energy-efficient housing for all resident groups. However, the communication document further acknowledges the current realities of energy poverty, with 34 million Europeans unable to keep their dwelling adequately warm [2].

Globally, energy poverty is the absence of sufficient choice in accessing adequate, affordable, reliable, high-quality, safe, and environmentally benign energy services to support economic and human development [3]. Within the framework of the EU, the definition is further centred around the costs of energy services and their affordability [4]. Several studies have identified links between energy poverty and socioeconomic, demographic, and ethnic characteristics, such as low-income levels [5,6], living in woman-lead households [6,7], and being a private renter [8,6]. Furthermore, overall higher rates of energy poverty have been observed in Mediterranean, Central, and Eastern European countries [6]. Even though the research on energy poverty within the context of energy expenditure dates to the end of the 1970 s [9]and was popularised in the context of the United Kingdom by beginning of the 1990 s [10], the price shock effects of the COVID-19 pandemic and Russia's war on Ukraine led to the need for more work done to ensure access of the vulnerable population to essential energy services [11].

In Estonia, social welfare policies are the principal approach to tackling energy poverty [12]. This leads to a reduced focus on energy poverty within the realm of the national energy policy, as observed in the revised National Energy and Climate Plan. However, the European Union's Emissions Trading System (EU ETS) transition to include the residential building sector can further accelerate energy poverty if the impact-reducing instruments' design does not consider existing inequalities [13]. Thus, assessing the existing inequalities in energy consumption in different groups of residents and housing market segments is important to determine suitable policy actions to address the changes

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in the EU ETS to reduce the number of energy-poor households and contribute to more equity in access to energy-efficient housing.

The paper aims to evaluate the existing differences in energy costs and consumption based on the housing and population characteristics in Tallinn, Estonia. Since less than 30 % of the residential buildings in the city have valid energy performance certificates, the study first and foremost aims to define and apply a machine-learning-based methodology for building energy performance predictions based on different open-access datasets, including remote sensing data, to assess the total energy consumption for the entire housing stock. Thereafter the multivariate analysis will be applied to estimate the causal relationships between the prediction variable based on the generated data and the building and neighbourhood level variables on housing and population characteristics. Even though several studies have been produced on the use of machine learning for building energy performance, the examples of the application of the modelled outputs in the context of urban studies and energy consumption inequality assessments are very limited. This makes this research unique and novel regarding methodology and its implementation. Additionally, similar approaches have not been applied in the context of Estonia and other European transition countries, considering the limitation of existing data availability. The study's research questions include: (1) What is the most suitable machine learning algorithm for the building energy performance prediction based on the existing data limitations? (2) What spatial and housing segment patterns can be observed in the variations of the energy costs across the city of Tallinn? (3) What neighbourhood-level population characteristics are associated with the dwelling energy cost variability?

2. Literature review

2.1. Urban energy inequalities

In many instances, the facilitation of energy transition was connected to increased inequalities, particularly as the activities are inaccessible to low-income households [14,5,15]. Grossman [15] notes that renovation activities are often associated with higher housing expenditures, threatening the notions of affordable housing and potentially requiring economisation on other vital human activities. The notion is strengthened further by Albrecht and Hamels [14], who found that approximately half of the homeowners in the Flemish region of Belgium cannot afford the renovation measures to achieve the 2050 climate policy goals. The disparities in the affordability of energy transition are highlighted even further by the existing funding schemes [5,16,17]. In Italy, the overrepresentation of wealthier households in using energy efficiency subsidies was observed [5]. To a similar extent, the Estonian building renovation grant programme allocation that was in use until recently followed regional development indicators, particularly local real estate prices, further exacerbating the existing regional inequalities [16].

Sustainable development of the urban environment has been linked to gentrification and displacement [18,19]. Quinton & Nesbitt [20] state that although many terms of gentrification caused by sustainable development exist, they represent different notions of intervention in the built environments associated with climate impact mitigation or adaptation. The work of Bouzarovski et al. [21] shows that the potential energy savings from building renovation activities can become a form of financial accumulation strategy benefiting owners, housing investors, and developers, driving gentrification further. Similar notions of resilience-driven gentrification within the communities of coastal areas affected by hurricanes are observed [22]. However, the notions of gentrification can also stem from the interactions between energy infrastructure, residents, and other forms of energy users [23,24,25]. Libertson et al. [25] describe energy gentrification as a form of gentrification where the competition over land is substituted with competition over energy. This is further observed in the context of ever-increasing reliance on data centres for high-tech services, the development of which was linked to displacement [24].

2.2. Machine learning for building energy performance prediction

A review by Seyedzadeh et al. [26] defines the machine learning approaches to building energy performance estimation as one of the four primary solution typologies. Furthermore, the review describes three main techniques for supervised machine learning: artificial neural networks (ANN), support vector machines (SVM), and Gaussian distribution regression models [26]. In addition, other studies have noted ensemble-based methods, such as random forest (RF) and eXtreme Gradient Boosting (XGBoost), to be suitable for the building energy performance calculation [27,28,29]. Within the unsupervised machine learning methods, k-mean and hierarchical clustering have been applied within the domain of building energy performance [26]. The following section will describe examples of different applications based on the overall diversity of the machine learning application for building energy modelling.

Tsoka et al. [30] and Aruta et al. [31] applied the artificial neural network (ANN) structures to facilitate classification and regression analysis of building energy performance. As a method, ANN represents a series of connected basic computing units for local information transmission, resembling how biological neurons operate [32]. In the structures of the ANN, the three primary layers are the input layer, the hidden layer, and the output layer, together with associated activation and transfer functions [30]. In the Tsoka et al. [30] study, the ANN was applied to building energy class label prediction based on the residential building energy performance certificates in the Lombardy region of Italy, which included input data on building characteristics, environmental conditions, and social factors. As the primary indicators, the study applied accuracy, sensitivity, and precision, with the two final models achieving 93.10 % accuracy for pre-2015 and 89.62 % for the post-2015 energy performance certificate methodology [30]. Aruta et al. [31] applied ANN for the time-series building energy consumption prediction for a net-zero-energy building based on the external dry bulb temperature and global solar radiation. The initial building energy performance is modelled in EnergyPlus and DesignBuilder software, where the different climatic zone conditions are modelled for the same building [31]. The study concluded that even with the study's limitations, the calculation time of the developed prediction algorithm is an important development to optimise computationally intensive modelling performed in traditional building energy performance modelling tools [31].

Support vector machines (SVMs) are a method of distribution-free learning for non-linear dependencies between the vector inputs and the dependent variable [33]. Dong et al. [34] applied the SVM algorithm to the four buildings in the Central Business District of Singapore to generate longitudinal annual energy consumption based on the building energy surveys performed between 1996–1998 and 2000–2001. As model performance indicators, the study applied mean squared error, mean squared error of scaled value, percentage error, and coefficient of variance based on root means squared error (RMSE) [34]. The final mean squared error varied between 0.73 and 0.14 based on the building for which the SVM model was developed, with the errors ranging between −2.72 % and 3.44 % [34]. However, the authors note that the potential performance of the algorithm can be attributed to the small data pool with fewer abnormalities [34]. Paudel et al. [35] also applied SVM for the energy load prediction based on the existing energy consumption data, climatic variables, occupancy profiles, and building operating conditions data for a residential building in France adhering to low-energy building standards. The TRNSYS software was further used to generate the building thermal data [35]. Furthermore, the model performance was evaluated based on coefficient of determination (R^2) and RMSE [35]. The study concluded that developing the relevant submodels for the different time periods can improve the model performance compared to applying all the available data [35].

Ensemble algorithms are a set of algorithms whose individual decisions are combined through weighted or unweighted voting as a way

of performance improvement [36]. Random forests (RF) represent one of the main ensemble methods for classification and regression tasks, where many decision trees are generated with additional randomness integrated to reduce overfitting [37]. Pham et al. [28] developed a short-term energy consumption prediction model based on five buildings' hourly energy consumption data using a random tree, M5 model trees, and RF algorithms. The study used mean absolute error (MAE), RMSE, mean absolute percentage error, and synthesis index as the performance indicators to evaluate and compare hourly, 12-hour, and 24-hour predictions [28]. The study concluded that applying the RF model yielded the most accurate prediction results [28]. Tsanas & Xifara [29] have also applied the RF model to predict the heating and cooling loads of the 768 developed building models based on the differences in building forms, glazing areas and their distribution, and building orientation. Mean squared error, mean relative error, and MAE were used to evaluate the model performance [29]. The study confirmed the suitability of RF algorithms for predicting heating and cooling loads with low deviations from the ground truth, which, in this instance, were the modelled energy loads [29].

In addition to the RF algorithms, different gradient-boosted algorithms have been applied to building energy modelling and predictions [27,38]. Gradient-boosted algorithms are trained in an additive manner with smaller trees compared to RF [39,38]. Seyedzadeh et al. [38] applied a gradient-boosted regression trees algorithm for the 4900 non-domestic buildings in the United Kingdom. The model performance metrics used the RMSE, MAE, and R2 [38]. The study notes the importance of feature engineering and extraction when applying the gradient-boosted regression trees [38]. XGBoost is an example of a highly efficient gradient-boosted algorithm due to the possibilities of parallel and out-of-core computations [39]. A study by Ali et al. [27] utilised different machine-learning-based algorithms, including XGBoost and RF, and end-use demand segregation methods to predict the energy performance of urban residential buildings. Within the study, a set of building archetypes for building energy performance prediction was created for Ireland, where different ensemble-based methods were cross-compared based on the performance of R^2, MAE, and RMSE [27]. XGBoost, together with the end-user demand segregation, was identified as the most suitable method for large-scale residential building energy performance modelling; however, as the method was tested on the synthetic data, the authors stress the data quality and availability as the primary determinants of the approach application in other contexts. [27].

Unsupervised machine learning methods involve constructing decision boundaries based on unlabelled data through clustering [40]. Seyedzadeh et al. [26] define two main typologies of clustering used in building energy consumption prediction as k-means and hierarchical clustering. The k-means clustering algorithm finds locally optimal solutions by minimising the potential clustering error based on the predefined number of cluster centres [41]. Talei et al. [42] describe the application of k-means clustering to identify the inefficient use of heating, ventilation, and air conditioning (HVAC) systems based on hourly energy consumption data. Through k-means clustering, the study identified a potential 6 % energy consumption reduction for the HVAC systems [42]. In Gao & Malkawi's [43] study, k-means clustering was also applied to a dataset of non-residential buildings in the United States, which included different building and energy equipment variables, to model energy use intensity. RMSE was used as the primary indicator of the model performance [43]. The study identified four clusters of buildings based on energy use intensity with significant variations in the actual values, the predicted values based on clustering, and the Energy Star methodology [43]. Instead of single data partitioning, hierarchical clustering develops a hierarchy of partitions, generally represented by a tree structure [44]. Ma et al. [45] applied hierarchical clustering for building energy performance assessment based on identifying the heating and electricity consumption load profiles. The study used data on heat energy and electricity consumption for a three-year period in a higher education building in Norway, identifying unique energy usage behaviours, particularly for heat energy [45].

3. Methodology

3.1. Study area

Tallinn was founded in the 13th century, with the rapid development of the city beginning in the 1860 s and continuing with industrialisation. During the Soviet Union occupation, the city expanded, and the population grew significantly due to external migration by mostly ethnically Russians, Belarussians, and Ukrainians [46]. Large housing estates such as Mustamäe, Väike-Õismäe (located in the inner part of Haabersti city district), and Lasnamäe were built from the 1960 s to the 1990 s to meet the housing demand (see Fig. 1). During this period, the socioeconomic differentiation of housing was low. In contrast, the ethnic differentiation was high between the housing types and districts, and the ethnically non-Estonian population was much more represented in the serial housing due to the housing allocation policies [46].

The post-socialist housing situation in Tallinn has been shaped by privatisation, the rise of private agents in property development, and suburbanisation [47]. The economic restructuring and the neoliberalization trends in the housing market have increased occupational and income disparities [48]. Large-scale social degradation tendencies were observed in the Tallinn housing estates, such as Lasnamäe and Mustamäe, particularly in the neighbourhoods with the highest concentrations of Russian-speaking minorities [49]. The inner city, comprising most neighbourhoods in the districts of Kesklinn, Põhja-Tallinn and Kristiine (see Fig. 1), on the other hand, has been restructured both economically and socially. The seaside areas have become desirable locations for real estate investors, showing the fastest increase in property developments and prices over the last 15 years, largely replacing existing residents through renewal of housing stock and gentrification [50]. Studies also reveal that the distinct social trajectories take place in different neighbourhoods of the inner city, where the reproduction of existing social diversity takes place, especially in the neighbourhoods inherited from the socialist era, in parallel with mild or more rapid gentrification and socio-economic upscaling [51]. Nõmme, Haabersti, and Pirita mostly represent the low-scale suburban neighbourhoods, but also have infills of high-rise small-scale quarters.

3.2. Data

Table 1 summarises the primary datasets used for the study. The analysis of the energy consumption of buildings is performed based on the building dataset from the Estonian topographic database provided by the Estonian Land Board and the energy performance certificate dataset managed by the Ministry of Climate of the Republic of Estonia. Both datasets were used for the extent of Tallinn city for March 2024 due to land surface temperature data availability. Building topographical data included variables such as year of building construction, building height, total projected building surface, and building purpose. Only a subset of data with all the variables mentioned and 'residential building' being the purpose of the were included in the study. To obtain the 10-metre resolution land surface temperature (LST), the open-access algorithm developed by Onačillová et al. [52] was used, which downscaled Landsat 8 and Landsat 9 thermal infrared images by using land cover metrics calculated for Sentinel-2 images. The LST dataset was generated on March 6, 2024.

As described in Table 1, for the occupational, demographic, and ethnic data, the 2021 Population Census data pre-aggregated to the level of statistical neighbourhoods was used. These statistical neighbourhoods represent morphologically coherent neighbourhoods, which follow street and building structures, historical developments, and local identity patterns [48]. Tallinn is divided into 230 statistical neighbourhoods with an average of two thousand residents. These

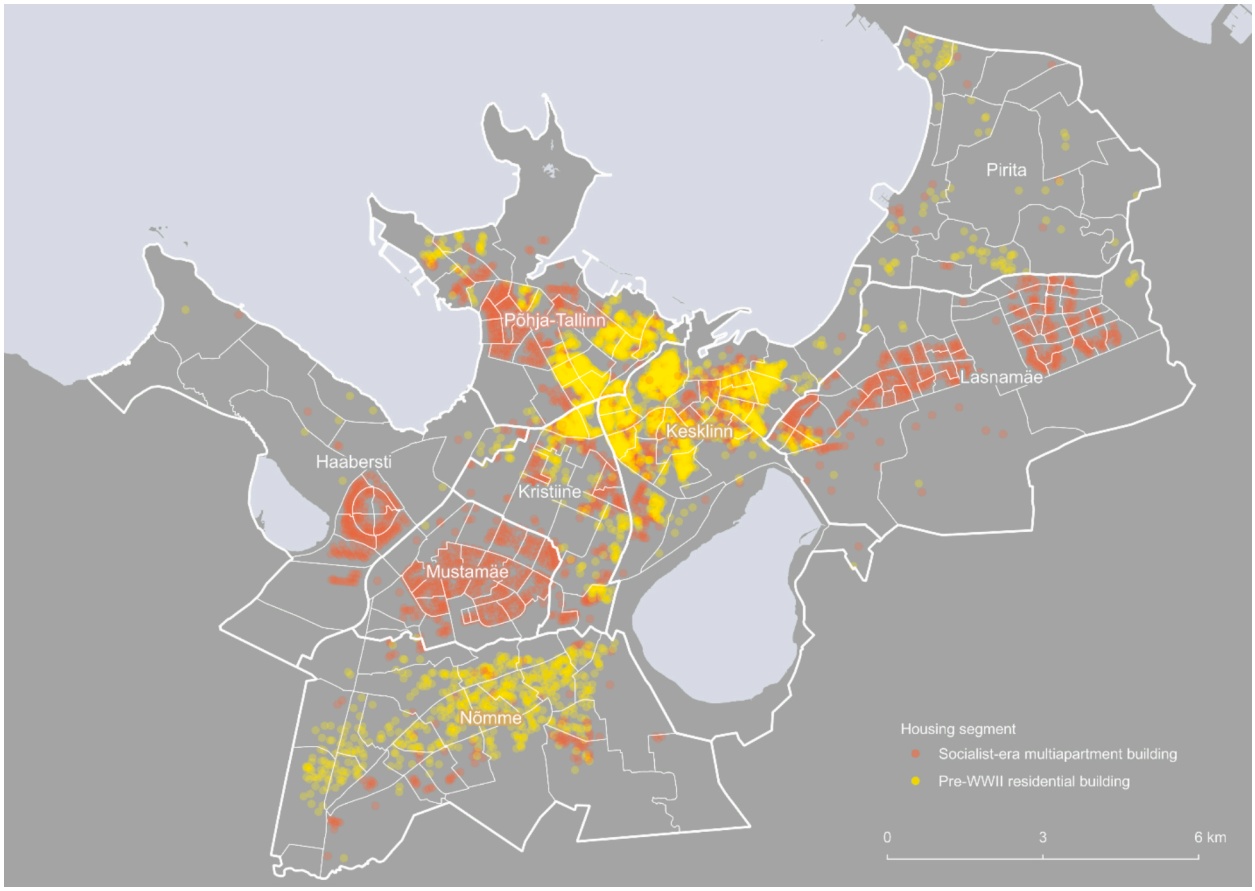


Fig. 1. Tallinn city districts and distribution of housing segments.

Table 1
Summary of the datasets used in the study.

Dataset	Type	Described time	Variables used
Building topographical data	V	05.11.2024	Building code, year of building construction, building height (m), total projected building surface (m^2), building purpose
Energy performance certificates	A	05.11.2024	Building code, energy label, total annual energy consumption (kWh/m^2)
Land surface temperature	R	06.03.2024	Land surface temperature (°C)
Building renovation grants	A	31.12.2023	Building code, date
2021 Population Census	V	31.12.2021	Number of individuals with Estonian, Russian, and other mother tongues Number of workers with ISCO 1–2, ISCO 3–4, and ISCO 5–9 occupational statuses Number of individuals aged 0–14, 15–65, and above 65 years

Note: Type denotes the dataset typology: 'V' stands for a vector dataset, 'R' stands for a raster dataset, and 'A' stands for an attribute tabular dataset.

neighbourhoods are relatively homogeneous regarding housing structures and serve as suitable bases for assessing locational (dis)advantages. Neighbourhoods with significant over-representation of ISCO 1–2 categories, which include managers and highly qualified specialists, can be found primarily in the central districts, inner city, and low-scale suburbs. In contrast, the locations of the blue-collar workers are over-represented in socialist high-rise neighbourhoods, but also since the housing fabric is spatially quite mixed the micro-level segregation takes place in the inner city and some lower-scale suburbs, so that instead of large-scale homogeneous districts of low and high social status categories the cityscape looks more like a patchwork of different socio-economic groups. The variability of the occupational groups and individuals based on the mother tongue is described in Fig. 2. Whereas the pre-WWII housing is mainly located in the inner city and the suburban area in the South, the high-rise socialist-era prefabricated housing neighbourhoods stand out as more homogeneous urban districts (see Fig. 1). The declared mother tongue is used as the ethnic variable in the study, dividing the population into three groups: Estonian speakers as natives, Russian speakers, and others. The population of Tallinn's neighbourhoods was divided further between three categories: people below the age of 15, people between the ages of 15 and 65, and people above 65 years of age.

Furthermore, Table 2 summarises the primary additional coefficients used in the study. As the energy performance certificates in Estonia describe the total energy demand associated with building energy use [53], electricity prices for February 2024 and district heating prices for August 2024 are used in the study, as the data reflected the prices of March 2024. Additionally, two district heating providers were identified as the only registered district heating companies within the Estonian Competition Authority operating in Tallinn [54]. Thus, the energy prices of both district heating providers are represented. Additional variables used in the study are the floor height and average dwelling size, which are used to estimate the number of building floors based on the building height and standardise the dwelling size for the total energy cost calculation for multiapartment buildings.

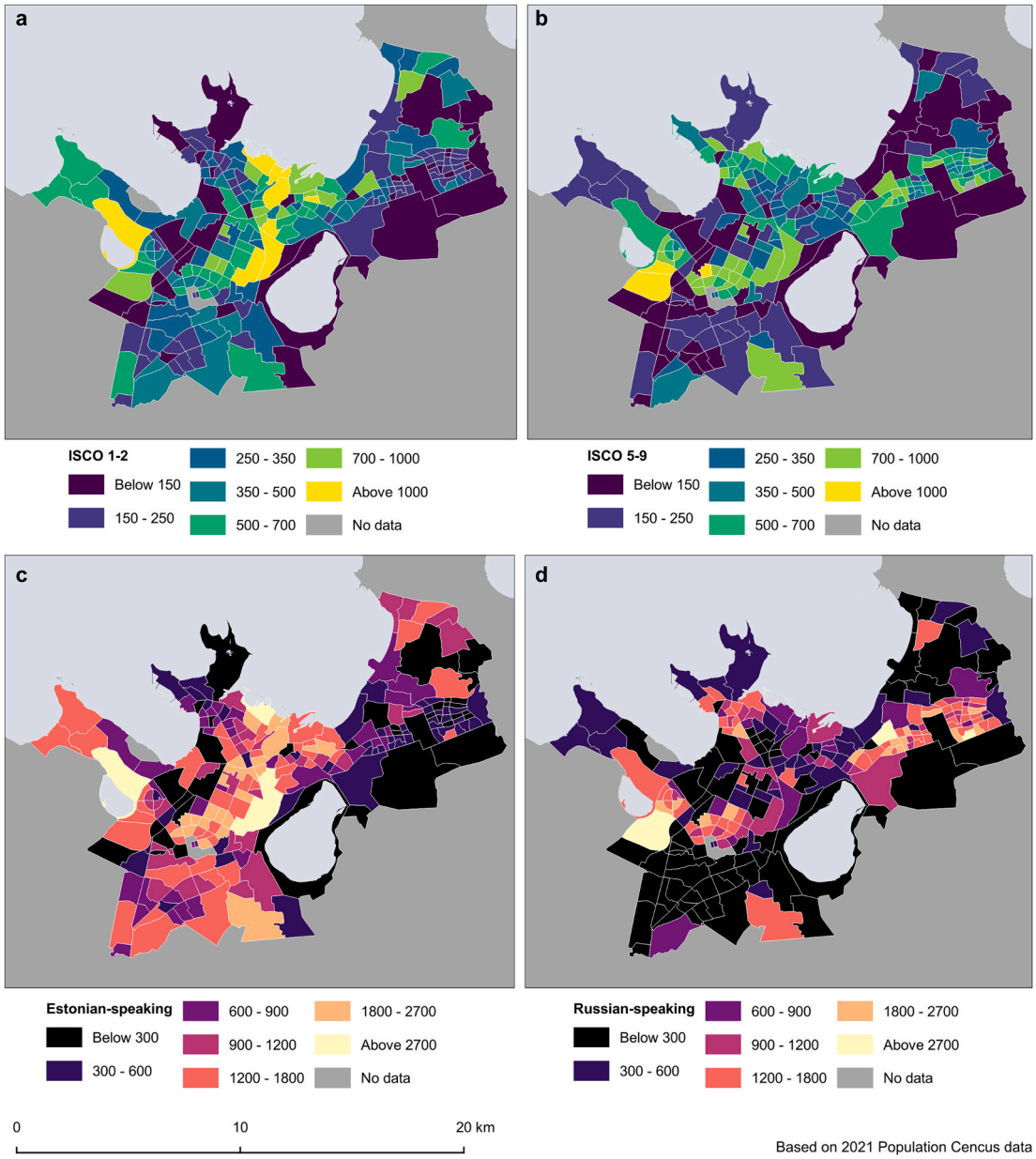


Fig. 2. Distribution of the number of individuals in the neighbourhoods of Tallinn. **Note:** (a) with high occupational status; (b) with low occupational status; (c) with Estonian as their mother tongue; (d) with Russian as their mother tongue.

4. Methods

The workflow of the study was designed in R, a statistical computational and graphics system, to increase further replicability of the developed framework. The overall methodological framework is described in Fig. 3, including model selection and application for Tallinn.

The work with the spatial vector datasets in the study was performed with the *sf* package in R [58,59]. Additional work with the raster files was performed by using the *terra* package [60]. The first step of the study was to sample LST for all the buildings. To do so, a mean value for the LST was calculated for the raster pixel centroid points within the boundaries of the building vectors. After the sampling, the EPC and building datasets were merged, creating a dataset for training and testing the machine learning algorithm. To integrate spatiality in the predictive analysis, the coordinates of the centroid points of buildings were used as additional predictive features. *Equation (1)* describes the input variables for the developed machine learning algorithm. The building volume was calculated by multiplying the projection area and height.

Table 2
Additional calculation coefficients used in the study.

Coefficient	Value	Unit	Source
Floor height	2.80	m	[55]
Average dwelling size	60.00	m^2	[56]
Electricity price	75.52	EUR/MWh	[57]
Heat energy price (1)	109.28	EUR/MWh	[54]
Heat energy price (2)	75.58	EUR/MWh	[54]

Note: The number in the heat energy price coefficient description refers to the district heating service provider: '1' indicates WTC Tallinn, and '2' indicates Utilitas Tallinn.

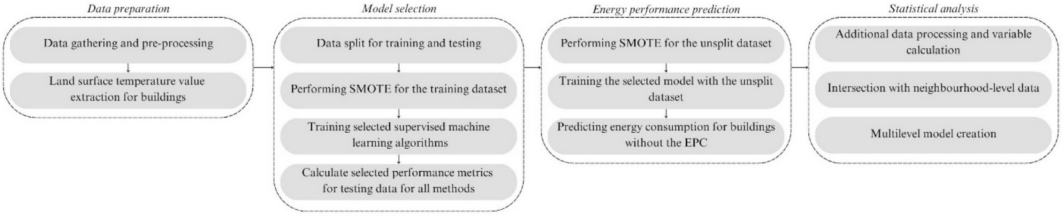


Fig. 3. Methodological framework of the study.

$$Y_{Tallinn} \sim (YC;\ Long;\ Lat;\ LST;\ S;\ H;\ V) \quad (1)$$

where

Y – predicted energy variable,
YC – year of building construction,
Long – the centroid points' longitude,
Lat – the centroid points' latitude,
LST – mean land surface temperature (°C),
S – building projection area (m^2),
H – building height (m),
V – building volume (m^3).

The model selection process performed 90 % training and 10 % testing data split. As the building energy efficiency labels were unequally distributed, the synthetic minority over-sampling technique (SMOTE) was used to introduce synthetic data to improve the data distribution between the classes. As a method, SMOTE performs a minority class oversampling within the line segments between the designated number of nearest neighbours of this class, later randomly choosing the required number of synthetic data points to minimise the differences between the class distribution [61]. In this instance, SMOTE was applied for training data only with the *smotefamily* package to mitigate the unequal data distribution [62]. No augmentations, including SMOTE, were performed for the testing dataset. Repeated cross-validation was performed during the model training to optimise the hyperparameters with the *caret* package.

4.1. Model selection process

Ten models were compared in the study based on the selected parameters, which are summarised in Table 3. As the methods represent classification and regression, only the buildings with energy labels and total energy consumption values were selected before the training and testing data split. To evaluate and compare performance between the different models, the classified energy classes were transformed into the associated total energy consumption values based on the definitions of the Ministry of Economic Affairs and Communications of the Republic of Estonia (Majandus- ja taristuministri, 2015).

Table 3
Summaries of the selected models.

Model	Abbreviation	Type of model	Package	Reference
Naive Bayes	NB	Classification	*klaR*	[63]
Support Vector Machine with Class Weights	SVM	Classification	*kernlab*	[64]
k-Nearest Neighbour	KNN	Classification	Base R	[65]
Random Forest	RF	Classification, Regression	*randomForest*	[66]
eXtreme Gradient Boosting	XGBoost	Classification, Regression	*xgboost, plyr*	[67,68]
Linear Regression	LR	Regression	Base R	[65]
General Additive Model	GAM	Regression	*mgcv*	[69]
Bayesian Regularized Neural Networks	BRNN	Regression	*brnn*	[70]

After training the selected classification and regression models, the prediction and cross-comparison of the results are based on the selected indicators calculated for the entire dataset. As the distribution of the errors varied across the different models used in the study, a set of different metrics was used. These include MAE, RMSE, and mean absolute deviation (MAD). RMSE is an optimal model performance metric for the normally distributed errors, whereas the MAE has been described as optimal for Laplacian errors [71]. Furthermore, to introduce a more robust model performance evaluation metric, MAD was introduced for error evaluation, whose distribution deviates from the normal [71]. The model was selected based on the comparative assessments of all three metrics together with the error distribution curves to determine the most suitable indicator for the assessment of the models. *Equations* (2), (3), and (4) describe the calculation of the three indicators used in the study:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|EC_{actual} - EC_{predicted}\right| \quad (2)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(EC_{actual} - EC_{predicted}\right)^2} \quad (3)$$

$$MAD = 1.483{*}median_{i=1}^{n}\left|EC_{predicted} - median(EC_{actual})\right| \quad (4)$$

where

EC_{actual}– actual building energy consumption, kWh/m^2,
$EC_{predicted}$– predicted building energy consumption, kWh/m2.

The final model applied all the available data for residential buildings without the testing and training data split. SMOTE was applied to the entire dataset to decrease the underrepresentation of building energy classes. The developed model was used to predict the missing final energy consumption data for residential buildings in Tallinn and the annual energy costs for the buildings on the dwelling level based on the building's heated area.

4.2. Statistical analysis

The predicted building-level data were used with the neighbourhood-level occupational, demographic, and ethnic data to develop a multilevel model assessing the energy costs. Multilevel or hierarchical modelling represents an established modelling framework in the social sciences to simultaneously study group-level and individual-level effects [72]. The model was implemented using the *lme4* R package to develop and analyse mixed-effect models [73]. The statistical model is summarised in *Equation (5)*. Every neighbourhood-level population variable group had an associated reference group excluded from the multilevel model. It included the Estonian-speaking population for the language categories, workers with high occupational status for the occupational groups, and people aged between 15 and 65 for the age groups. The absolute population counts within the categories were divided by the total number of people within the

categories to further work with proportional data. Furthermore, these variables were scaled, together with the building age and heated area. Log transformation was also performed for the total heated area before the scaling.

$$EC_{nb} = \beta_0 + \beta_1 L_RU_n + \beta_2 L_{OTHn} + \beta_3 O_M_n + \beta_4 O_L_n + \beta_5 A_C_n + \beta_6 A_E_n + \gamma_n + \beta_7 BA_b + \beta_8 I(RG)_b + \beta_9 I(MF)_b + \beta_{10} BHA_b + \varepsilon_{nb}, \quad (5)$$

where

n – neighbourhood-level variable.

b – building-level variable.

EC_{nb}– total annual energy costs for a dwelling in a building.

$\beta_0 - \beta_{10}$– estimate values.

L_RU_n, L_OTH_n – primary languages used in households (Russian, Other).

O_M_n, O_L_n – working individuals occupational status groups (Medium, Low).

A_C_n, A_E_n – individuals by age groups (C: 0–14, E: 65 +).

BA_b– building age.

$I(RG)_b$– KredEx renovation grant allocation for the building.

$I(MF)_b$– multifamily building.

BHA_b– building's heated area.

γ_n– random neighbourhood effect.

ε_{nb}– random building effect.

After developing the multi-level model, the variance inflation factor (VIF) was used to determine the multicollinearity between the variables using the *performance* package [74]. In addition, the random neighbourhood effects from the multilevel model were plotted for the additional assessment of the unaccounted variabilities in energy costs and their potential links to the spatial distribution.

5. Results

5.1. Model selection

The performance of the testing metrics for the model selection is summarised in Table 4. Based on the initial assumption of the study, different algorithms performed best for different metrics. The lowest MAE value was observed for the regression RF algorithm, whereas the KNN classification algorithm performed the best based on the RMSE. However, based on MAD, the RF and XGBoost performed the best. The variation of the results based on the different metrics required plotting the error curves for every algorithm to determine the most suitable for predicting building energy performance for the residential building stock of Tallinn.

Fig. 4 represents the results of error distribution plotting and error interval assessment ordered based on the median absolute error value. The distribution of all model prediction errors is not Gaussian and Laplacian; thus, MAD as the indicative value will be prioritised. Furthermore, comparing the median absolute error shows slight performance improvements when applying the RF classification algorithm for the prediction compared to XGBoost. Thus, the final model for predicting building energy consumption is RF, which is used for classifying building energy labels. Since the total number of buildings with an active energy label is higher than the total number of buildings with the value of total energy consumption, the predictive model training includes more input data than the training dataset for the model selection.

Table 4
Summary of the model selection performance indicators.

Model	Type	MAE	RMSE	MAD
NB	Classification	51.57	73.84	49.68
SVM	Classification	54.57	65.28	49.68
KNN	Classification	47.28	31.00	49.68
RF	Classification	37.69	54.25	24.47
XGBoost	Classification	37.88	54.36	24.47
RF	Regression	34.96	284.61	40.22
XGBoost	Regression	39.38	311.83	45.84
LR	Regression	48.51	340.64	58.12
GAM	Regression	43.91	335.94	49.70
BRNN	Regression	42.89	33.42	49.70

5.2. Multilevel model

The RF classification model was trained on all available and valid energy label data to predict the building energy performance. After developing the final model and predicting the energy labels for the missing buildings, the total number of buildings with energy labels increased from 3836 to 22841, representing 82.47 % of Tallinn's total residential building stock. The results of the predictions were aggregated and mapped to the level of small neighbourhoods. The results can be observed in Fig. 5. On building energy consumption per square metre, the neighbourhoods of Kesklinn and Põhja-Tallinn – i.e. the districts with the over-representation of historical WW2 housing – have substantially higher energy consumption than the other neighbourhoods in the city. However, the outcomes on modelling the annual energy costs when considering the size of the dwelling's heated area and differentiating between the detached housing and apartment buildings revealed that the pre-WW2 single-family dominated neighbourhoods of Nõmme had significantly higher annual energy costs. The spatial analysis further shows that the overall structure of energy costs and energy consumption varies by location. However, the socialist high-rise apartment districts, and especially Lasnamäe, which was built the latest, mainly from the 1970 s to the 1990 s, stand out as having the lowest levels of energy costs per dwelling.

The data was used to develop the multilevel model for building-level characteristics and neighbourhood-level population composition data. The results are summarised in Table 5. The building level characteristics confirm the following observations: the building age is positively associated with the building energy consumption, multifamily buildings have significantly lower energy costs per dwelling compared to detached dwellings, and total heated area on the building level increases the energy costs per dwelling. Additionally, the acquisition of renovation grants is associated with lower energy costs for multi-apartment buildings.

Four statistically significant phenomena occur based on the multilevel model for the neighbourhood-level variables. Compared to the Estonian-speaking population in the neighbourhood, both Russian-speaking and populations with other mother tongues experience statistically significantly lower annual energy costs on the dwelling level. These groups tend to be overrepresented in the socialist-era apartment buildings. Furthermore, the increase in the share of in middle occupational groups, who are more often also found in socialist-era blocks of flats in the Tallinn case, is also associated with a decrease in the annual energy costs compared to the high occupational groups. It is further important to note that compared to the shares of high occupational groups on the neighbourhood level, the variations in the proportions of low occupational groups did not yield statistically significant coefficients. Within the age groups, compared to the shares of working populations, the increase in the share of the elderly population on the neighbourhood level is associated with the decrease in dwelling-level energy costs. Finally, the model's VIF values show low multicollinearity levels across all the model variables.

After modelling, the random neighbourhood effects were plotted in Fig. 6. Overall, the random neighbourhood-level effect has several emerging patterns: almost all the neighbourhoods in the Mustamäe district show a negative random effect on the dwelling-level energy costs. This can be explained by the high share of homogeneous socialist-era multiapartment housing found in the district. To a similar extent, some neighbourhoods of Haabersti with housing structures similar to those of Mustamäe and most neighbourhoods in Lasnamäe show this

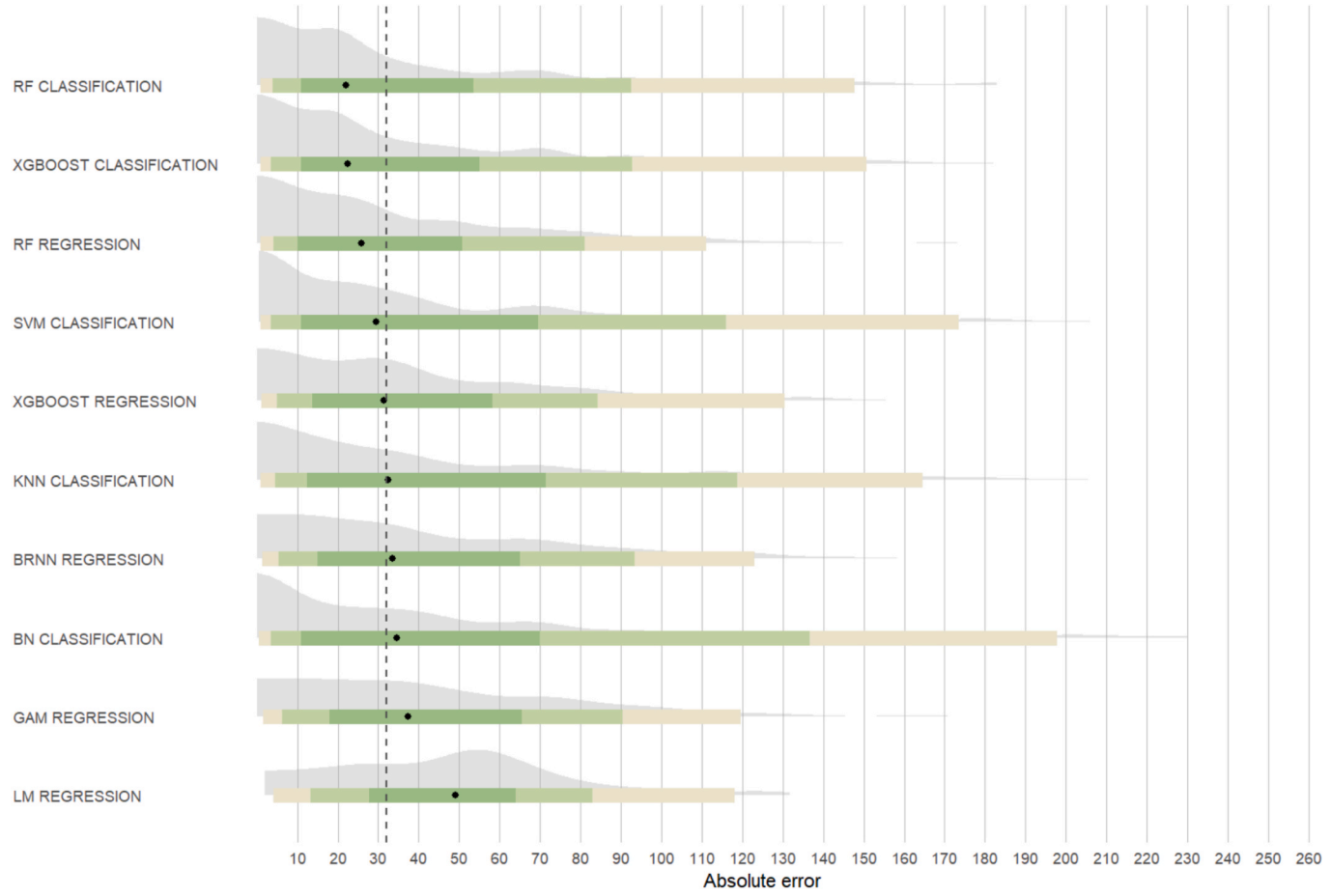


Fig. 4. The absolute error distribution curves and error distribution confidence intervals. **Note:** The dot denotes the median absolute error for the predicted energy consumption on testing data. The green colour represents the 50% confidence interval range, pale green represents the 80% confidence interval range, and beige represents 95% of the confidence interval range. The stroked line represents the global median absolute error range between all the models. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

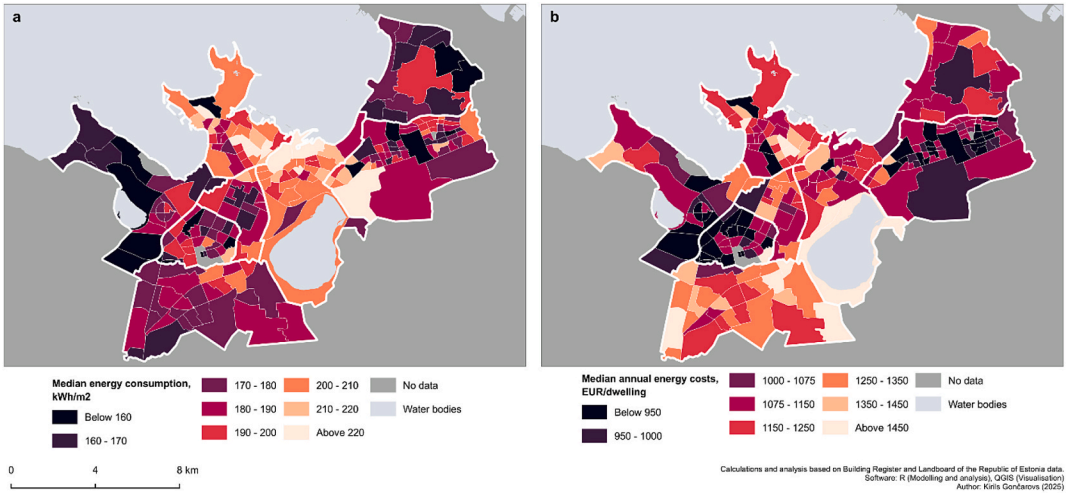


Fig. 5. The map of Tallinn neighbourhoods (a) median annual energy consumption per square metre; (b) median annual energy costs per dwelling.

negative random effect on energy costs in dwellings. However, the neighbourhoods with higher shares of detached housing in Haabersti also show the highest positive random effects, similarly to the neighbourhoods of Pirita. The other districts of Tallinn have less unified patterns of random neighbourhood effect distribution.

6. Discussion

The study successfully proved that in the absence of energy certificates, it is possible to estimate energy consumption using a machine-learning-based methodology utilising remote sensing and publicly available building data. The study's main innovation was to use a

Table 5
The results of the multilevel model.

	Estimate	Std. Error	p-value	Significance	VIF
Intercept	1861.27	11.37	$< 2.00 * 10^{-16}$	***	–
Building age	78.05	0.12	$< 2.00 * 10^{-16}$	***	1.06
Renovation grants (YES = 1)	−55.71	29.62	0.06	.	1.02
Multiapartment building (YES = 1)	−835.42	11.05	$< 2.00 * 10^{-16}$	***	1.77
Total heated area	163.41	6.85	$< 2.00 * 10^{-16}$	***	2.15
Russian-speaking population (n)	−58.71	11.41	$2.67*10^{-4}$	***	3.31
Population with other mother tongues (n)	−39.29	10.66	$5.28*10^{-7}$	***	3.14
Middle occupational groups (n)	−18.03	7.31	0.01	*	1.41
Low occupational groups (n)	−7.68	14.01	0.58	–	4.90
People aged 0–14 (n)	−22.19	14.25	0.12	–	3.94
People above 65 years of age (n)	−46.59	12.54	$2.67*10^{-4}$	**	3.14

Note: Values with (n) describe neighbourhood-level proportional data. Significance codes are denoted as "***" for the p-values between 0 and $1*10^{-3}$; "**" for the p-values between $1*10^{-3}$ and 0.01; "*" for the values between 0.01 and 0.05; '.' for the p-values between 0.05 and 0.10; and '-' for p-values above 0.10.

combination of two approaches – application of different machine-learning algorithms and statistical multilevel modelling – to study the variations of energy costs based on the building data and neighbourhood-level population characteristics. Furthermore, the application of remote sensing-derived data expanded upon the available datasets for urban-level building energy performance predictions and improved the model performance. Only the open-access datasets were used to develop the building energy performance machine learning algorithm, thus enabling the replication of the approach within the contexts of similar data availabilities on the national level in the follow-up research. The study further showcased the methodological possibilities of linking dwelling-level energy cost and consumption patterns with the neighbourhood population data, enabling the study of energy inequality in the case of limited access to more granular data levels.

The results of the multilevel models indicate that differences in energy costs and consumption can largely be explained by the spatial segmentation of the housing market, whereas implementing energy renovations proved to have a further positive influence on the reduction of energy use and costs. The historical buildings in the gentrifying city quarters (buildings emanating from the pre-WWII period) stand out with the highest energy consumption and cost. These are mainly located in the central and inner-city areas. Nevertheless, even more than the inner-city located apartment buildings, which in most cases have been refurbished, lately also better insulated and located in the central district heating networks, the low-rise suburban pre-WWII district Nõmme stood out as the highest energy-consuming neighbourhood per dwelling. This leads to one of the key findings that, next to the large-scale socialist apartment buildings, which tend to be the main beneficiaries of the renovation energy-efficiency grant schemes, the historical smaller-scale buildings are with actually in more profound need of energy improvements, and where the gap between the actual and potential costs after the potential renovation activities is the largest. Whereas the socio-economic groups exposed to higher energy use in the inner-city areas are higher socio-economic groups and working-age individuals, in the garden city of Nõmme, elderly groups are also represented. In both locations, the Estonian-speaking population tends to be over-represented.

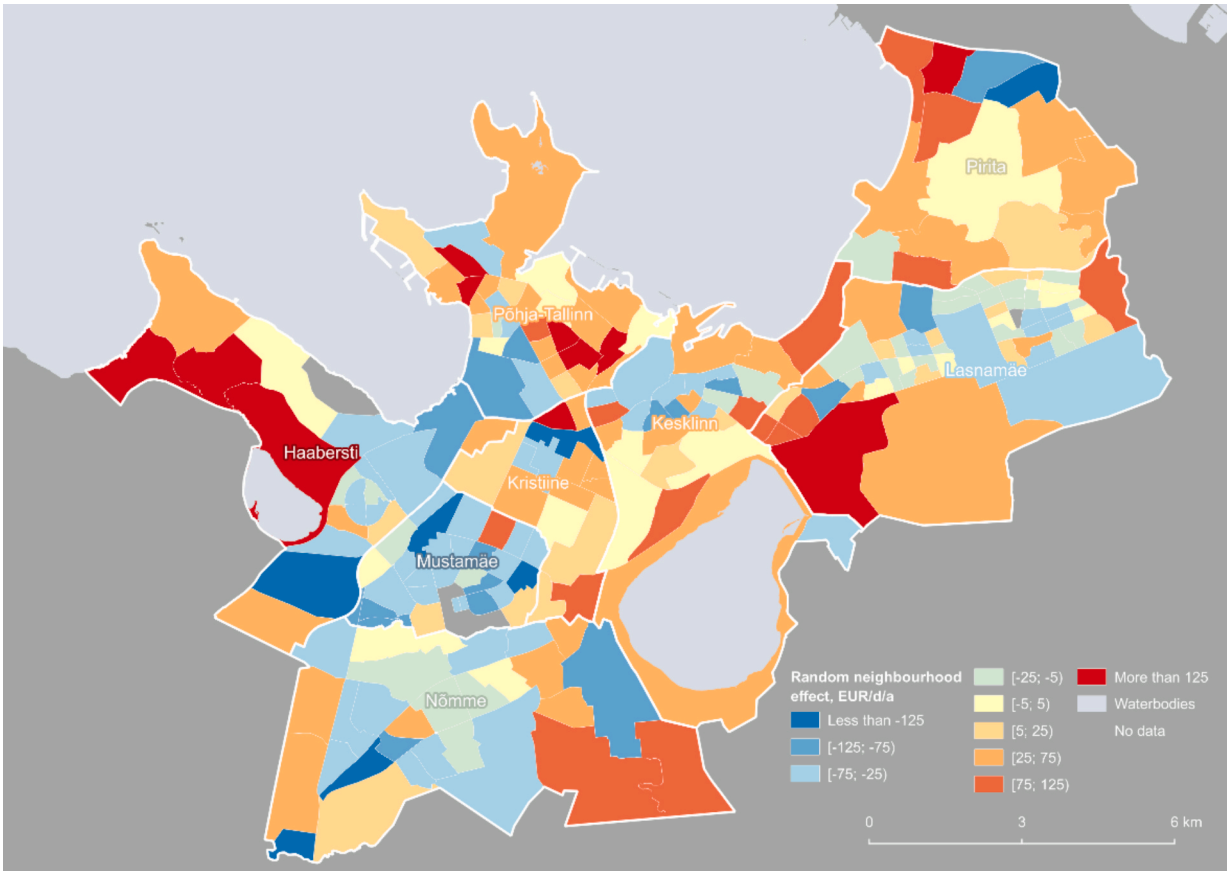


Fig. 6. Random neighbourhood-level effect in the multilevel model.

Socialist-era housing estates erected during the '70 s to the '90 s stand out with moderate energy consumption per square metre and cost per dwelling. However, the city landscape in the latter neighbourhoods tends to be the poorest and downgraded morally and physically. Hence, the need for physical restructuring and renovation is a hotly debated issue there despite the fact that energy consumption per se is not as pressing an issue in those areas. Our energy cost analysis shows that the potential for savings from future regeneration due to energy conservation is limited. The socialist housing estates thus exhibit a housing market segment that, given its current status of energy consumption, is likely to hinder the further development of deep renovation.

The housing segmentation intersects with residential distribution. The socio-spatial, ethno-spatial and age-distribution patterns considerably overlap, with a strong concentration of working and middle professions and Russian-speaking groups in the socialist housing estates [49]. Those neighbourhoods also tend to stand out among the ageing population. Non-Estonian background, middle social status, and older age indicate increased chances of living in moderately energy-consuming and lower energy-cost housing. The intersection between the overall lower renovation capacity, as in the case of the elderly, and the lower pressing need for energy-retrofitting leads to the reduced likelihood of the housing estates becoming successful renovation test labs in the near future. Our results reveal that Estonia's background, higher socio-economic status, and working age are related to higher energy costs and consumption. Since people of Estonian background more often live in single-family homes and historical buildings, these are the housing types and neighbourhoods with the highest energy costs. Thus, Tallinn's spatial segregation of housing types and uneven distribution of population characteristics further amplify inequalities in energy consumption and costs.

The population age group analysis revealed the overall negative trend between the neighbourhood-level proportion of the elderly population and dwelling-level energy costs compared to working adults. The results of lower annual energy costs can be attributed to the lower residential mobility levels of the elderly populations living in the housing estates in the post-socialist countries [75], as the population group is highly represented in the neighbourhoods of Lasnamäe and Mustamäe. Furthermore, many of the neighbourhoods in these two districts also have high negative values for the random effect. However, it is important to note that, in this instance, the lower energy costs in the neighbourhoods with higher proportions of the elderly population do not minimise the potential levels of energy poverty experienced by elderly households.

The results of the study to some extent contradict with the existing studies on energy poverty carried out in other urban contexts which for example have revealed the higher overall energy costs in case of minority households [76], or the higher socio-economic categories and gentrifiers being the main winners of energy transition resulting in access to energy-networks, overall deduction of energy-costs, and lower-income groups detected as the most exposed to the high energy costs [5,77,25]. Somewhat disputing those results, our study outputs point to the fact that the groups who are often considered as vulnerable – non-natives, elderly, or middle-occupations tend to be exposed to lower energy costs in the case of the city of Tallinn. However, the housing and residential mosaic in Tallinn is heterogeneous, and making generalisations may ignore small-scale 'hidden' patterns where for example vulnerable groups such as poorer households or elderly settle in unrenovated buildings which tend to consume high level of energy, especially within the fabric of older housing stock in neighbourhoods which are undergoing social and physical upgrading and gentrification. Therefore, when considering housing energy-retrofitting needs, households' socio-economic status and capability to engage in renovation activities should be assessed, and appropriate measures applied.

The study results are important in indicating new socio-spatial divides caused by the different exposure to energy consumption and energy poverty, and when considering suitable financial impact reduction measures to alleviate the risks of increased energy poverty, and to support energy renovation. However, the study had several limitations derived from its design framework. First, the available building-level parameters were limited, which resulted in the generalisation of some criteria, such as applying the same mean dwelling size for the energy costs in apartment buildings and using unified energy costs for Tallinn. This potentially resulted in reduced cost variabilities across the different buildings in the city. The follow-up study could apply higher resolution of population data to confirm the observed phenomena and a more comprehensive analysis of trends in energy costs between the different groups. Furthermore, the potential integration of the income data can allow for applying the developed machine-learning method of building energy performance prediction for evaluating energy poverty through cost-based methods.

7. Conclusions

The study aimed to identify the housing, location-specific, and household-specific criteria determining the energy cost variation in Tallinn, Estonia. The comparison between ten machine learning algorithms for classification and regression tasks was performed using publicly available building data and energy performance certificates. The random forest classification algorithm was used based on the best performance for the selected metrics and applied to increase the total number of buildings with predicted energy consumption values. A multilevel model was developed to evaluate the variation of energy costs between the different population groups, considering the housing characteristics. Finally, the random neighbourhood effect was detected by applying the multilevel model.

The multilevel analysis revealed the important interdependence between the spatial housing segmentation and energy consumption, with the historical dwellings' neighbourhoods emanating from the pre-socialist era, and the detached housing units, especially, consuming most energy. Newly built dwellings and socialist high-rise estates tended to exhibit more moderate energy consumption, referring to the lowest motivation for their residents to engage in energy retrofitting activities. Since the socio-economic status of people living in socialist housing stock is skewed towards lower and middle-social status groups, complemented by the overrepresentation of the elderly, the capacity to energy-retrofit buildings is relatively limited in those areas. On the other hand, the results show that building retrofitting further reduces energy use and cost.

In terms of the population characteristics related to the decreased energy costs in Tallinn, the results were as follows:

1. The increase in the proportion of the elderly population compared to the working-age population on the neighbourhood level was associated with decreased dwelling energy costs.
2. The increase in the proportion of Russian or other language-speaking populations compared to the Estonian-speaking population on the neighbourhood level was associated with decreased dwelling energy costs.
3. The increase in the proportion of workers in middle occupational groups compared to the high occupational groups on the neighbourhood level was associated with decreased dwelling energy costs.

The lack of statistically significant trends in energy cost patterns for other groups, such as individuals with low occupational status, reveals the overall heterogeneity of energy consumption trends in case of Tallinn, Estonia. This can partly be explained by the patchwork-type spatial layout of the city in terms of housing types, with rather small-scale, distinctive neighbourhoods instead of large, homogeneous districts, except the socialist high-rise and dense residential district of Lasnamäe.

The characteristics identified in this study are crucial in considering the implementation of the EU ETS system expansion to the residential building sector and facilitation of just energy transition, as the basis of

assessing the most impacted population and housing segments in Tallinn. These results can further be used to create more targeted actions to reduce the potential negative socio-economic impacts on the population groups involved.

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CRediT authorship contribution statement

Kirils Gončarovs: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anneli Kährik:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Kirils Goncarovs reports financial support was provided by University of Tartu Institute of Ecology and Earth Sciences. Anneli Kahrik reports financial support was provided by University of Tartu Institute of Ecology and Earth Sciences. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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